Rishik Thammisetty

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Summary

Pre-final year Electrical and Electronics Engineering undergraduate at NIT Andhra Pradesh with research interests in Computer Vision, Deep Learning, Medical Image Analysis, and AI-driven Electronic Design Automation. Currently working as a Research Intern at IIT Bhilai, developing novel deep learning architectures for image inpainting with a focus on efficient FPGA implementation and real-time computer vision applications.

Education

National Institute of Technology Andhra Pradesh — Tadepalligudem, India
B.Tech. in Electrical and Electronics Engineering; CGPA: 8.35/10 (Current) — ***Aug 2025 – Present***

Minor Degree in AI and Robotics — ***Aug 2025 – Present***

Technical Skills

Languages: C, Python, MATLAB, Verilog

ML Frameworks & Tools: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV, SHAP, CUDA

Core Domains: Machine Learning, Deep Learning, Computer Vision, Medical Image Analysis, Image Processing, VLSI Design, EDA

Other Tools: MS Excel,Power BI,Tableau Xilinx VIVADO, Web Development, DSA, Embedded Systems

Professional Experience

Indian Institute of Technology (IIT) Bhilai — **May 2026 – Present**
Research Intern

- Designed a novel attention-enhanced deep learning architecture for image inpainting to reconstruct severely degraded image regions with improved perceptual quality.
- Investigated convolutional and transformer-inspired modules for improving reconstruction accuracy and computational efficiency.
- EvaluatING models using PSNR, SSIM, and LPIPS while exploring FPGA-based deployment for real-time inference.

Deloitte Australia – Virtual Experience (Forage) — **April 2026 – May 2026**
Data Analytics & Forensic Technology Virtual Intern

- Performed data analysis using Tableau to create interactive dashboards and derive actionable business insights.
- Analyzed complex datasets to identify patterns and trends, directly supporting data-driven decision-making processes.
- Gained practical exposure to industry-standard workflows utilized in consulting and forensic technology projects.

Academic Projects

Advanced Deep Generative Image Inpainting | Github — **May 2026 – Present**

- Designed and developed a novel deep learning architecture for image inpainting to reconstruct missing or corrupted image regions with high visual fidelity, utilizing convolutional and attention-based mechanisms.
- Conducted comprehensive experimental evaluations using standard computer vision metrics, optimizing the architecture for computational efficiency and real-time hardware acceleration via FPGA implementations.

Machine Learning-Assisted Static Timing Analysis (ML-STA) | Github — **May 2026 – Present**

- Developing an ML-assisted Static Timing Analysis framework to predict critical path delays, identify timing bottlenecks, and detect violations in digital circuits to accelerate design closure.
- Designing an end-to-end circuit analysis pipeline to extract high-dimensional timing features from gate-level netlists and timing graphs, reducing overall analysis complexity within EDA workflows.

AI-Based EV Battery Health Prediction System | Github — **Jan 2026 – Apr 2026**

- Developed an integrated ML-based system to predict State of Health (SoH) using State of Charge (SoC), Remaining Useful Life (RUL), and automated anomaly detection as supporting features.
- Applied robust data preprocessing, feature engineering, and regression algorithms to improve battery degradation prediction accuracy under dynamically varying operating conditions.

Certifications & Coursework

- **Certifications**
  - **NVIDIA**: Building Transformer-Based Natural Language Processing Applications
  - **Google Cloud / Simplilearn**: Introduction to Large Language Models
  - **NPTEL**: Design and Analysis of VLSI Subsystems (Elite Certified)

Research Interests

Computer Vision, Medical Image Analysis, Deep Learning, Explainable AI, EDA, FPGA-based AI Acceleration